

Mitigation of Common Mode Voltage in 2-Level 3-Phase Voltage Source Inverters

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Abstract—This paper introduces a novel method to mitigate the common mode voltage (CMV) issue in 2-level 3-phase (2L-3Ph) voltage-source inverters (VSIs). Up to now, several solutions have been proposed to mitigate CMV, such as using passive/active filters, introducing additional switches, and utilizing specially designed pulse-width modulation (PWM) methods. Since the other solutions increase the cost and complexity, specially designed PWM solutions have become attractive and more promising. However, existing literature generally states that PWM solutions primarily reduce the high-frequency switching components of CMV, though CMV due to dv/dt remains because of switching instants. This study proposes a novel PWM method to eliminate CMV from dv/dt by synchronizing the PWMs of VSI's legs. This synchronization method is infeasible for direct utilization in conventional sinusoidal PWM (SPWM); therefore, this study introduces a third-harmonic square-wave injection and an adaptive carrier phase shift method to make it possible. In an ideal case, the proposed method completely removes dv/dt , but, in practice, a small impact of dv/dt might still be observed because of the effects of the non-idealities of the switches and gate driving circuits. The practicality of the proposed method is demonstrated through an in-depth analysis of both analytical and simulation results. Finally, the effectiveness and feasibility of the proposed method were validated using an experimental setup with SiC devices.

Index Terms—Common mode voltage, voltage source inverter, motor drive, pulse width modulation synchronization, sinusoidal pulse width modulation

I. INTRODUCTION

Wide-band gap (WBG) devices such as silicon carbide (SiC) and gallium nitride (GaN) have become game-changers in motor drive systems owing to their superior properties, which provide numerous benefits, such as greater compactness and higher efficiency [1]–[3]. However, WBG devices must be fast-switched to fully utilize these benefits, but this results in high dv/dt at switching instances, which generates common mode voltage (CMV) [4]–[7]. This CMV might result in bearing current and shaft voltage, leading to premature bearing breakdown, which is one of the most frequent failure types in motors [8]–[10]. Therefore, researchers and industry professionals have investigated various CMV mitigation methods to prevent such failures, which are classified into three main groups:

connection-side, motor-side, and inverter-side solutions [11], [12].

First, connection-side solutions involve adding filters between the inverter and motor, such as common-mode chokes or shortened paths using resonant structures [13], [14]. These methods effectively attenuate the common-mode current through the bearing or parasitic capacitances of the motor. However, the passive elements utilized can be bulky and costly. Second, motor-side solutions use new insulation materials in bearings and windings or grounded shaft [6], [7], [15]–[17]. Such approaches directly mitigate bearing currents or avoid breaking the winding insulation; however, their effectiveness is prone to motor design. Moreover, new types of materials may have undistinguished mechanical properties and may be expensive. Lastly, inverter-side solutions involve the implementation of new topologies or the development of innovative modulation strategies [18]–[28]. Although new topologies, such as multi-level or 2-level topologies with additional switches, increase the cost and require complex control strategies, innovative modulation techniques offer the advantage of streamlined integration, requiring minimal additional space and noncomplex modifications. Therefore, this study focuses on modulation-based solutions.

In [29], a modified space vector modulation (SVM) was proposed, which decreases the amount and amplitude of common-mode voltage by reducing the number of transitions to six. In [30], the switching patterns were updated to reduce the CMV for two PWM algorithms: near-state PWM and the modified form of the active zero-state PWM. Here, the proposed method performs well for both low and high modulation indices. In [31], a CMV reduction method was developed for carrier-based sinusoidal PWMs (CB-SPWM) where carrier phase shifts between the legs of the 3ph-VSI (0° , 120° , and 240°) are introduced. In [32], the CMV reduction for CB-SPWM was improved by introducing the sampling time adaptive carrier phase shift method, which yields carrier phase shifts 0° or 180° with respect to the reference voltage of the legs. However, the methods discussed above reduce the amount and magnitude of CMV, and they cannot eliminate dv/dt . Other modulation methods to eliminate dv/dt have been proposed in a dual-inverter structure for 3-phases or a single-inverter structure for multiphase inverters. In these modulation methods, the main idea is to synchronize the PWMs to cancel the rising and falling dv/dt values of the legs. However, this synchronization procedure is infeasible when using conventional sinusoidal SPWM or SVM in a single inverter with a 3ph-2L system. This study introduces a new modulation technique based on

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CB-SPWM to make 3ph-2L motor drives feasible for dv/dt cancelation based on PWM synchronization. A third-harmonic square wave is injected in the proposed modulation technique, and the variable carrier phase shifts that are updated for each switching period are introduced. In this way, complete dv/dt cancelation is achieved in an ideal case. The main contribution of the study can be listed as follows:

- 1) A new modulation method based on conventional SPWM is introduced to reduce CMV, allowing it to work alongside traditional drive-of-motor control.
- 2) This modulation method can be easily implemented without additional computational burden, making it suitable for use with off-the-shelf microcontrollers or FPGAs for motor control.
- 3) Although this method is designed for 2-level, 3-phase drive systems, it is not limited to them. It can be easily extended to multilevel, multiphase drives, making it a versatile solution.
- 4) By completely reducing the common-mode dv/dt , this method increases the lifetime of motor insulations and bearings.

The rest of this paper is organized as follows. Section II presents the system structure and analyzes the common-mode voltage in the 2 L-3Ph-VSIs. Section III presents the mathematical background of the mitigation method and explains the injection and adaptive carrier phase shifts. Section IV discusses practical issues in the implementation and provides a generic form for multiphase machines. Section V gives the simulation results to validate the analytical calculations and Section VI presents the experimental setup and validation. Finally, Section VII concludes the study.

II. SYSTEM STRUCTURE AND COMMON MODE VOLTAGE ANALYSIS

A. System Structure

Fig. 1 shows the traditional 2-level, 3-phase motor drive system, and a simplified common mode equivalent circuit diagram. These equivalent circuit parameters represent the stray capacitances between the windings and the cores of the electrical motor.

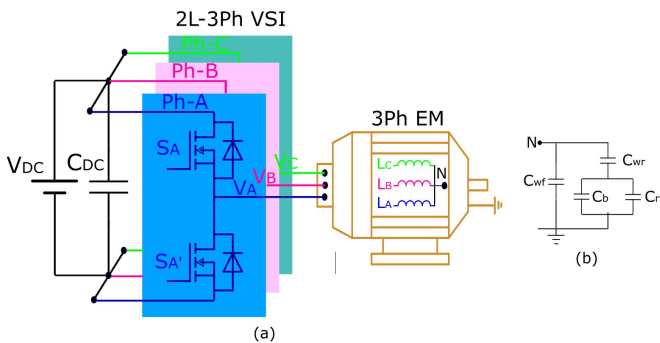


Fig. 1. a) The system structure of a conventional 2-level 3-phase motor drive system. b) Equivalent common mode circuit diagram.

C_{wf} , C_{wr} , C_{rf} , and C_b represent the stray capacitances between the stator winding and frame, stator winding and

rotor, rotor and frame, bearing and frame, respectively. These stray capacitances play a significant role in common-mode currents. Their influence becomes more noticeable at high frequencies because they provide low-impedance paths for the common mode currents to flow through. Therefore, if the motor-side parameters remain unchanged, reducing the high-frequency common-mode voltage at the inverter output may be the only solution to prevent failures caused by these common-mode currents.

B. Common Mode Voltage Analysis

Regarding the system shown in Fig. 1, the CMV is calculated by taking the average value of three-phase voltages as given by

$$V_{CM} = \frac{1}{3}(V_A + V_B + V_C) \quad (1)$$

where V_A , V_B , and V_C are phase-to-neutral voltages. Then, considering that the inverter is driven by carrier-based (CB) SPWM, the phase-to-neutral voltages are altered between $-V_{DC}/2$ and $+V_{DC}/2$ where V_{DC} is the pole-to-pole input voltage.

CB-SPWM can be produced by comparing sinusoidal references with triangular carrier signals. Conventionally, it is prevalent to use a shared single-carrier signal for each phase with 120° phase-shifted sinusoidal reference signals. In such a case, the carrier, reference signals, and switching pulses during a fundamental period are shown in Fig. 2.a and Fig. 2.b.

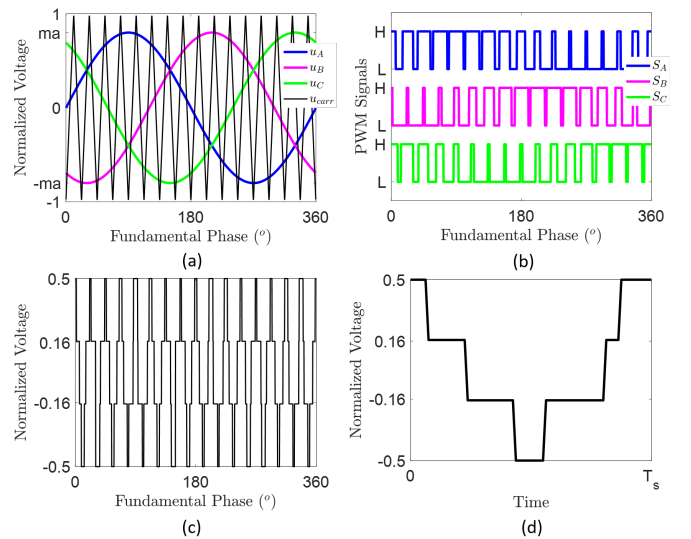


Fig. 2. The key waveforms of 2L-3Ph VSI. a) Reference and carrier signals of conventional SPWM. b) PWM signals for the high-side switch of each leg. c) Normalized CMV of the 2L-3Ph VSI driven by SPWM. d) Close-view of the normalized CMV for a single switching period.

Therefore, the CMV, which is calculated using (1) along with the fundamental and switching periods, is shown in Fig. 2.c and Fig. 2.d, respectively. It is observed that the common mode voltage (CMV) frequency is three times the switching frequency, and the switching transitions (dv/dt) occur six times within a switching period. In addition, the

CMV voltages cause current to flow through the stray capacitances, as represented in the common-mode equivalent circuit. Therefore, the impact of the CMV increases with both the switching frequency and switching speed of the MOSFETs. However, selecting the switching frequency and speed requires a holistic approach to optimize the overall system. For instance, the switching frequency is chosen based on the motor's requirements for differential current ripple and harmonic losses, whereas the switching speed is selected based on the inverter's electrical loss and thermal constraints. Consequently, utilizing new modulations would effectively mitigate CMV without altering the switching frequency and speed. This approach can significantly improve the motor drive system performance without introducing additional cost and complexity.

III. THE PROPOSED MITIGATION METHODOLGY

The proposed method investigates a single switching period, instead of a fundamental period. If CMV is eliminated within a switching period at each operating point of the SWPM, the overall CMV is eliminated. Fig. 3 illustrates carrier and reference signals, phase-to-neutral output voltages, and CMV during a single switching period for both a single carrier signal and multiple carrier signals with phase shifts, represented as $\phi_{cA} = 0$, $\phi_{cB} = \frac{t_B}{T_s} 2\pi$, $\phi_{cC} = \frac{t_C}{T_s} 2\pi$ where t_A is taken a reference and is selected zero, t_B , and t_C are time delays between the carrier signals.

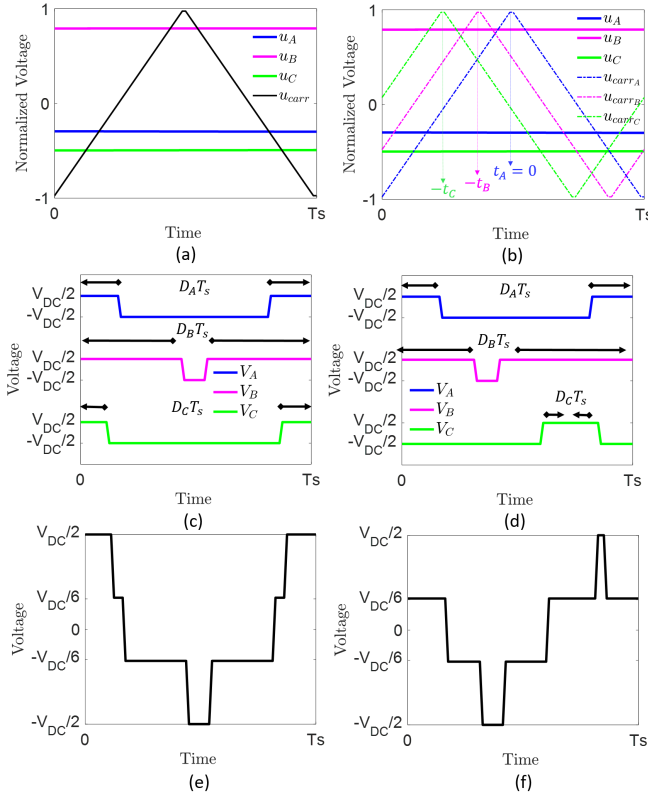


Fig. 3. Closeup waveforms for a single switching period. Carrier and reference voltages: a) Single-carrier SPWM b) Multi-carrier SPWM. Phase-to-neutral output voltages: c) Single-carrier SPWM d) Multi-carrier SPWM. Common mode voltages: e) Single-carrier SPWM f) Multi-carrier SPWM.

Although the duty cycles are given the same for both single-carrier and multiple-carrier SPWM, Fig. 3.e and 3.f show that the shapes of CMVs are different, so the key idea to eliminate CMV is to introduce multiple carriers with distinct phase shifts. These duty cycles within switching periods are determined by the sinusoidal reference signals which are given by

$$\begin{aligned} u_A &= m_a \sin(2\pi f_o t) \\ u_B &= m_a \sin(2\pi f_o t - \frac{2\pi}{3}) \\ u_C &= m_a \sin(2\pi f_o t + \frac{2\pi}{3}) \end{aligned} \quad (2)$$

where m_a is the modulation index, f_o is the fundamental frequency and t is time. Accordingly, the duty cycles can be calculated by

$$\begin{aligned} D_A &= \frac{1 + u_A}{2} \\ D_B &= \frac{1 + u_B}{2} \\ D_C &= \frac{1 + u_C}{2} \end{aligned} \quad (3)$$

Then, the phase voltages within a switching period can be reflected in a square wave with distinct duty cycles and phase shifts. Hence, the output voltages within a switching period (denoted as V_h) can be represented by a Fourier series, as given by

$$V_h = \sum_{n=-\infty}^{\infty} c_n e^{j2\pi n f_{sw} t} \quad (4)$$

where f_{sw} is the switching frequency, and c_n are the Fourier coefficients, which can be calculated with respect to the harmonic number as given by

$$c_n = \begin{cases} \frac{V_{DC}}{-j2\pi n} (e^{-jn(2\pi D + \phi_C)} - e^{-jn\phi_C}) & \text{if } n \neq 0 \\ \frac{V_{DC}}{2} D & \text{if } n = 0 \end{cases} \quad (5)$$

where D are duty cycles and ϕ_C are carrier phase shifts. In fine, to eliminate overall high-frequency CMV, the summation of harmonic components of the phases, and so the summation of c_n s, should be zero, as given by

$$\frac{1}{3} \sum_{ph} (A, B, C) \sum_{n=1}^{\infty} c_{nph} e^{j2\pi n f_{sw} t} = 0 \quad (6)$$

In this sum, the n^{th} harmonic Fourier coefficients for CMV are calculated by

$$c_{nCMV} = c_{nA} + c_{nB} + c_{nC} \quad (7)$$

The aim is to find carrier phase shift conditions that give c_{nCMV} zero in this sum concerning duty cycles that follow the reference signals. To find the conditions, the Fourier series of c_{nCMV} can be explicitly written by

$$\begin{aligned} c_{nCMV} = & \frac{V_{dc}}{-j2\pi n} \left(e^{-jn(2\pi D_A + \phi_{cA})} + e^{-jn(2\pi D_B + \phi_{cB})} + e^{-jn(2\pi D_C + \phi_{cC})} \right. \\ & \left. + e^{-jn\phi_{cA}} + e^{-jn\phi_{cB}} + e^{-jn\phi_{cC}} \right) \end{aligned} \quad (8)$$

For this equation to be valid, both sets of terms (the first set of exponentials involving duty cycles and phases, and the second set involving phases only) must nullify each other. Thus, the following equalities must hold for the method to be valid:

$$\begin{aligned} 2\pi D_A + \phi_{CA} &= \phi_{CB} \\ 2\pi D_B + \phi_{CB} &= \phi_{CC} \\ 2\pi D_C + \phi_{CC} &= \phi_{CA} \end{aligned} \quad (9)$$

Then, these equations are summed together, the single equation is held as given by

$$\begin{aligned} 2\pi D_A + \phi_{CA} + 2\pi D_B + \phi_{CB} + 2\pi D_C + \phi_{CC} \\ = \phi_{CB} + \phi_{CC} + \phi_{CA} \end{aligned} \quad (10)$$

To vanish all harmonic terms, this depends on the condition given by

$$2\pi(D_A + D_B + D_C) \equiv 0 \pmod{2\pi} \quad (11)$$

Accordingly, the sum of the duty cycles should be an integer (1,2,3, etc) to cancel all harmonic components. When the sum of duty cycles meets the integer condition, the required carrier phase shifts can be calculated regarding the duty cycles as given by

$$\begin{aligned} \phi_{cA} &= 0 \\ \phi_{cB} &= (D_A + D_B)\pi \\ \phi_{cC} &= (D_A + 2D_B + D_C)\pi \end{aligned} \quad (12)$$

where, for simplicity, assume that $\phi_{cA} = 0$. An exemplary demonstration of two distinct cases (the sum of duty cycles is 1 and the sum of duty cycles is 2) can be seen in Fig. 4. However, in conventional SPWM, the sum of the duty cycles

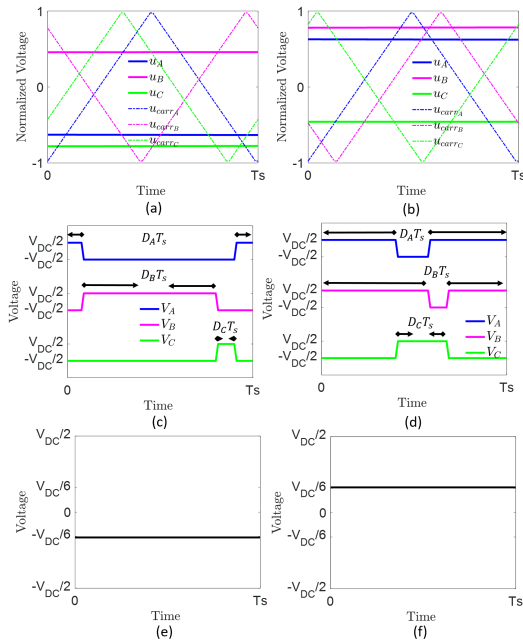


Fig. 4. Closeup waveforms for a single switching period. Carrier and reference voltages: a) $D_{sum} = 1$ b) $D_{sum} = 2$. Phase-to-neutral output voltages: c) $D_{sum} = 1$ d) $D_{sum} = 2$. Common mode voltages: e) $D_{sum} = 1$ f) $D_{sum} = 2$

is constant (3/2) for each operating point as calculated by

$$D_{sum} = \frac{1+u_A}{2} + \frac{1+u_B}{2} + \frac{1+u_C}{2} = \frac{3}{2} \quad (13)$$

The reference signals (u_A, u_B, u_C) create a positive sequence, so the sum is zero. Therefore, to cancel the CMV, the reference voltage should be updated to achieve the sum of the integer number. The first idea is to inject a DC signal (as a common mode) into the reference signal to hold the integer sum condition. Therefore, each reference signal can be subtracted or added by 1/3 to achieve the sum of duty cycles of 1 or 2. Fig. 5.a and 5.b show the positive and negative DC addition to the reference signals to achieve constant CMV. The required carrier phase shifts for DC-injected SPWM to cancel the CMV are illustrated in Fig. 5.g and 5.h. However, in this

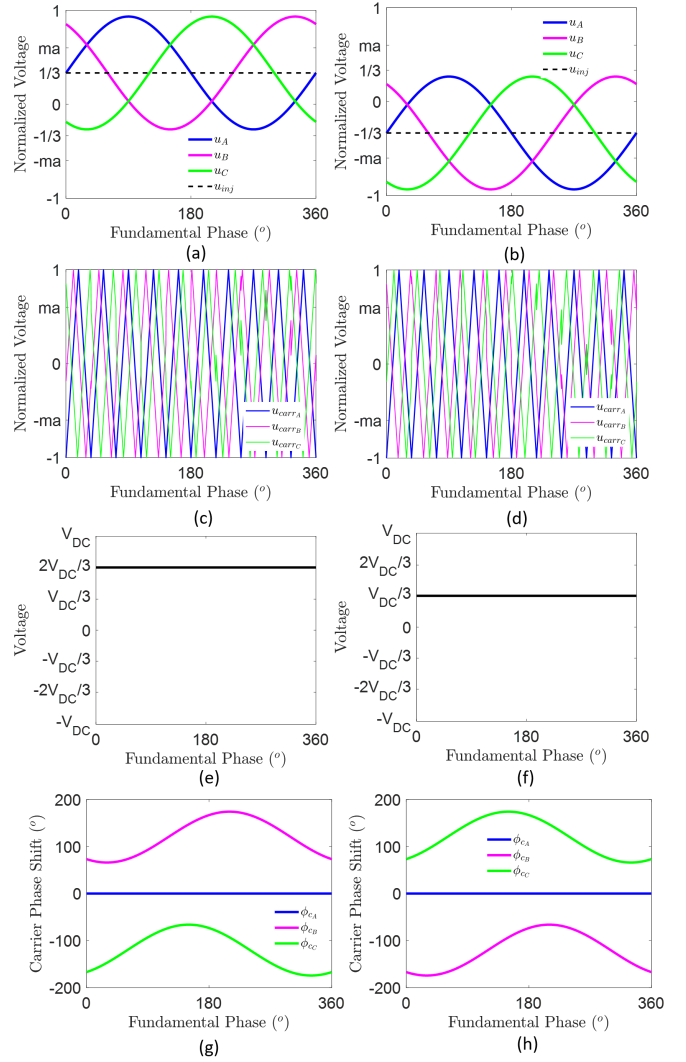


Fig. 5. Example DC injections with $m_a = 0.6$. Reference signals. a) +1/3 DC Injection. b) -1/3 DC Injection. Carrier signals. c) +1/3 DC Injection. d) -1/3 DC Injection. CMV. e) +1/3 DC Injection. f) -1/3 DC Injection. Carrier phase shift Values. g) +1/3 DC Injection. h) -1/3 DC Injection.

situation, the maximum modulation index is limited to 2/3 because of 1/3 DC injection. The modulation index limitation should be enhanced to increase DC-link voltage utilization. Therefore, the second idea is to inject a third harmonic square

wave that is like altering positive and negative DC injections periodically. Although the amplitude of the square wave is the same with DC injection (1/3), the maximum modulation index is augmented. The reference signals and injected square wave are illustrated in Fig 6.a and 6.b. As seen in 6.a, the

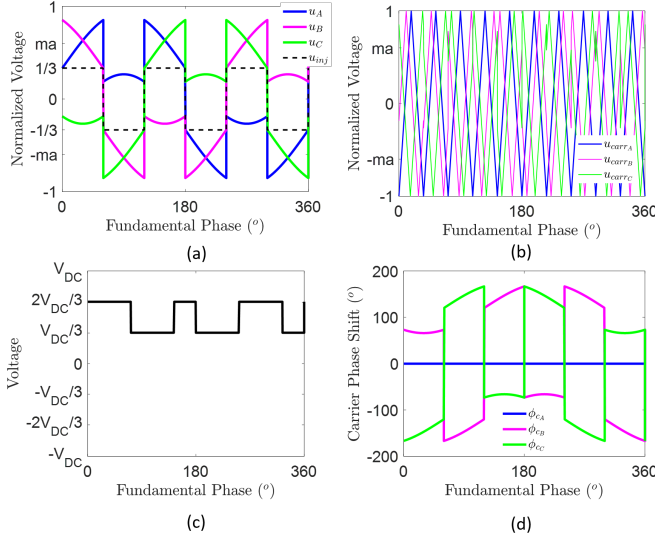


Fig. 6. Example square wave injection with $m_a = 0.6$. a) Reference signals. b) Carrier signals. c) CMV. d) Carrier phase shift Values.

maximum modulation index can be calculated at the 60° of phase-A. The maximum value at this angle should be 1, and so the maximum modulation index is found 0.77 as calculated by

$$\begin{aligned} m_a \sin(\pi/3) + (1/3) &= 1 \\ m_a &= \frac{1 - (1/3)}{\sin(\pi/3)} \\ m_a &= \frac{2/3}{0.866} = 0.77 \end{aligned} \quad (14)$$

CMV is observed in low-frequency because of injected signal, but as discussed before, the common mode currents become dominant in high-frequency, and low-frequency common mode voltage excitation is a common practice in motor drives to increase the DC-link utilization. Therefore, the effect of the third harmonic square wave injection becomes very low compared to high-frequency CMV. Consequently, the overall modulation strategy is shown in Fig. 7.

IV. DISCUSSION

A. Modulation Limitations

In the proposed method, the modulation index is limited to 0.77. This limitation affects performance parameters like efficiency and power density. However, the use of a low modulation index might be encountered in specialized applications like aerospace, where high-voltage DC-link systems are utilized, and lower AC output voltages are demanded because of machine design aspects [33], [34]. Consequently, the modulation index should be considered a critical design parameter. By configuring motor windings in either series or

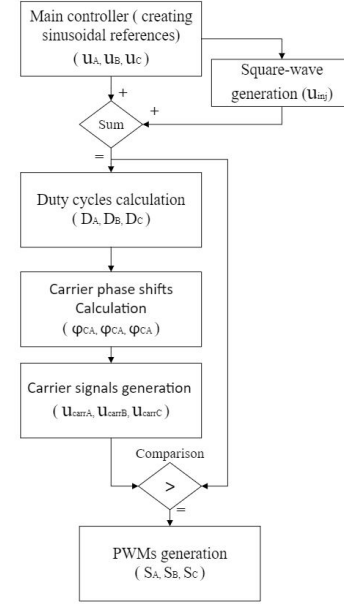


Fig. 7. Flow chart diagram of the proposed modulation technique. (The flow chart will be updated later !!!)

parallel, it is possible to achieve equivalent motor performance with a lower modulation index at the same DC-link voltage.

B. The Impact of the Non-idealities

First, the dv/dt depends on the operating conditions (phase currents), which means that the rising and falling edges of different legs might not compensate for each other. Even without precautions, a natural partial reduction in dv/dt still occurs. A more effective solution to solve dv/dt variations is to introduce variable gate resistance, which allows a constant dv/dt under all operating conditions. Second, the impact of dead time and gate driver delay mismatches, which change the phase shift calculations, must be accounted for. To overcome this problem, the gate driver delays can be precisely measured, and the dead times appropriately adjusted, incorporating these corrections into the phase calculations.

C. Generalization for Multi-phase VSIs

As derived in 11, the sum of duty cycles should be integer to be able to synchronize PWMs and cancel dv/dt . Even if we increase the number of phases, this condition does not change, and the sum of duty cycles for multi-phase inverters for SPWM can be calculated by

$$D_{sum} = \frac{\#Phases}{2} \quad (15)$$

First, for the even numbers of phases, the sum of the duty cycle becomes intrinsically integer, which means that there is no need for square wave injection. Therefore, the limitation of modulation disappears inherently. Second, for the odd numbers of phases, we still need square wave injection, but the amplitude of the injection becomes smaller while increasing the phase number, which means that the modulation limitation decreases. Consequently, the synchronization method can be

easily employed in multi-phase inverters, and increasing the phase number makes the drawback of modulation limitation lower.

V. SIMULATION RESULTS

I will add this section after getting experimental validation.

VI. EXPERIMENTAL VALIDATION

I will add this section using Imperix SiC modules, RL load, and the common mode parasitic circuit that I showed in the section II.

A. Test Setup

B. No-load Test

C. load Test

VII. CONCLUSION

This paper presented a novel PWM method to effectively eliminate common mode voltage (CMV) caused by dv/dt in 2-level 3-phase voltage source inverters. By synchronizing the PWM signals of the inverter legs, facilitated through third harmonic square wave injection and adaptive carrier phase shift, the proposed method mitigates the CMV issue. In this study, the comprehensive analysis of the modulation by Fourier series was given to determine the conditions to meet. The analytical calculations for the required phase shifts were conducted. It was observed that the proposed method also works under inherent non-idealities of switches and gate drive circuits. The practicality and robustness of the proposed technique were thoroughly validated through analytical models and experimental tests with SiC devices. The experimental results confirm that this method offers a promising approach for minimizing CMV in high-performance inverter systems without increasing system cost or complexity.

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